

MAYANK BANSAL

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SKILLS

- Data & Languages:** Python, NumPy, Pandas, Matplotlib, Seaborn, SQL, C++, Java, C
- Backend & Deployment:** Flask, FastAPI, REST APIs, Docker, Git, Render
- Machine Learning & AI:** Scikit-learn, TensorFlow, PyTorch, CNNs, Neural Networks, Feature Engineering, Model Evaluation, EDA, MLOps, Deep Learning, Computer Vision, NLP, Data Science
- Vector Search & RAG:** FAISS, Sentence Transformers, LangChain, RAG, ChromaDB, LangGraph
- Soft Skills:** Problem Solving, Leadership, Public Speaking

INTERNSHIP

- Machine Learning Intern | Qriocity Ventures | [Offer Letter](#)**
Jun 2026 - Jul 2026
- Developed deep learning models for medical image analysis using PyTorch, TIMM, and Computer Vision techniques.
 - Constructed and trained a U-Net (ResNet34) for pneumothorax segmentation (Dice Score: 0.78) and an EfficientNet-B4 system for multi-class chest disease classification (weighted F1: 0.91).

PROJECTS

- AI Career Copilot | [GitHub](#)**
Aug 2026 – Sep 2026
- Delivered an agentic RAG career assistant and ATS using LangGraph tool orchestration, with a human-in-the-loop workflow that pauses execution for interactive resume diff review before applying changes.
 - Designed a dual-provider LLM architecture (Groq cloud, Ollama local) with a ChromaDB semantic retrieval layer over sentence-transformer embeddings.
 - Achieved 90% Recall@4 and 0.83 MRR@4 on a 15-query benchmark, with Groq delivering 0.81s P50 latency.
 - Tech Stack** - Python, FastAPI, LangChain, LangGraph, RAG, ChromaDB, Groq, Ollama, Docker
- Multi-Hop Technical Debug Assistant | [GitHub](#)**
Jun 2026 – Jul 2026
- Architected a production-style Multi-Hop RAG assistant with metadata-aware chunking, semantic retrieval, persistent FAISS indexing, and multi-document search.
 - Configured a FastAPI application with live knowledge uploads, grounded LLM responses, source attribution, retrieval scores, and performance metrics.
 - Recorded ~800ms average response latency while delivering grounded responses with source attribution using Llama 3.3 via Groq.
 - Tech Stack** - FastAPI, FAISS, Sentence Transformers, Groq (Llama 3.3)
- Visual Search Engine | [GitHub](#)**
Feb 2026 – Mar 2026
- Built an image retrieval system using pretrained ResNet50 feature extractor generating 2048-dim embeddings and FAISS nearest-neighbour search for real-time visual similarity.
 - Engineered a FastAPI backend for image preprocessing, embedding generation, and vector search with a responsive upload-and-preview frontend.
 - Optimized the end-to-end pipeline to achieve approximately **150ms retrieval latency** during inference.
 - Tech Stack** - PyTorch, FAISS, FastAPI, JavaScript

CERTIFICATES

- LLM Engineer LaunchPad | [LPU](#) Jul 2026
- Complete Data Science & ML | [GeeksForGeeks](#) Nov 2025
- AI Tools Skill Up | [GeeksForGeeks](#) Aug 2025

ACHIEVEMENTS

- Published a technical book** — "Machine Learning Projects: A Practical Guide for B.Tech CSE Students," on Amazon Kindle and Paperback, covering 6 end-to-end ML projects across NLP, CNNs, Transfer Learning, and Audio ML | [Amazon](#)
- Published research paper** — "Mood2Mail: A Lightweight Real-Time Email Tone Analyzer Using Classical ML" on Zenodo
- Competitive programming** — 580+ problems on GeeksForGeeks (101-day streak, rank 159 / 39,000+ participants) and 190+ on LeetCode

EDUCATION

- Lovely Professional University** Punjab, India
 Bachelor of Technology - Computer Science and Engineering; **CGPA: 9.41** Aug 2024 - Present
- S.T. Joseph's International School** Haryana, India
 Senior Secondary (Class XII), [CBSE] | 93% May 2024